Sample Project

Title: Railway_Guages

First, we have to import required library. Then we have to import the CSV file into Pandas library.

Pandas is a powerful data analysis library providing flexible way to do data manipulation. There are many mathematical and scientific computational features can be leveraged using this library.

Matplotlib is a Python library used for data visualization it helps you create graphs and charts from data. It's one of the most popular tools in Python for plotting.

In order to load the CSV file using Pandas, we need to do the following step.

```
#import Required Libraries
import pandas as pd
from matplotlib import pyplot as plt

# Load Dataset
d = pd.read_csv("railway_gauges.csv")
```

Pandas returns as Data frame object. Using data frame object, we can see the top 5 rows using **head()** function and **tail()** function to display last 5 rows in a CSV file. In our above example, we have used **df.head()**, it displays first 5 rows in the loaded CSV. Here I am displaying first 5 & last 5 rows and columns.

Using head() it displays first 5 rows from dataset.

	A	B	C	D	E
1	Year	Broad Gau	Metre Gau	Narrow Ga	Total
2	1964-65	3521	2891	464	6876
3	1965-66	3603	2913	470	6986
4	1966-67	3615	2911	481	7007
5	1967-68	3601	2920	506	7027
6	1968-69	3603	2921	508	7032

Using tail() it displays last 5 rows from dataset.

46	2008-09	6185	642	203	7030
47	2009-10	6320	543	220	7083
48	2010-11	6419	494	220	7133
49	2011-12	6494	432	220	7146
50	2012-13	6577	379	216	7172

- Using Matplotlib we can plot the data flow in graphical representation format. Plotting bar graph for the above given years.

```
# plotting bar graph
ax = d.plot(x="Year", kind="bar")
```

Now, adding title and labels to the bar

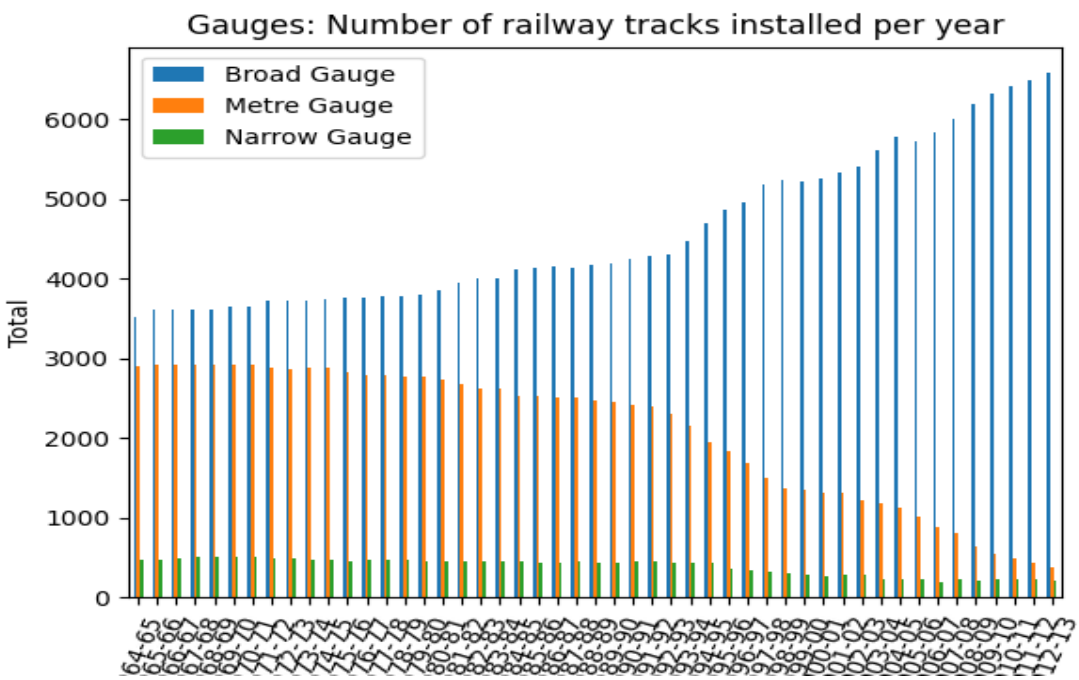
- **Title:** The title is the main heading of the graph. It tells what the whole graph is.
- **Label:** Labels describe the axes (sides of the graph).

Part	Meaning	Where it appears
Title	Overall name of graph	Top
X-label	Meaning of horizontal axis	Bottom
Y-label	Meaning of vertical axis	Left side

- Title = what the graph shows overall
- X-axis = Year data
- Y-axis = Total number of gauges

```
# Adding labels and title to the bar graph
plt.xticks(rotation=70)
plt.xlabel("Year")
plt.ylabel("Total")
plt.title("Gauges: Number of railway tracks installed per year")
```

- After finishing the required steps then we need to save the output figures and then display the output graphs.

```
# saving figure and dislaying
plt.savefig("rail_gauges.png")
plt.show()
```

- And the output graph look likes in below.